Transfer learning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Transfer learning before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Similarity and normalisation

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Sequence representations

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pooling and spatial information

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A source model may learn useful features before we have many labels for a new target task. **Transfer learning** reuses those learned features or parameters. One approach freezes the **feature extractor** and fits a new **output head**.

Fine tuning instead updates some or all of the reused parameters, usually together with the head.

Use a transparent stand-in for a source-trained extractor: f(x) = ReLU(x1 + x2), with supplied weights [1, 1] and bias 0. ReLU keeps a positive value and replaces a negative value with zero.

Here x1 and x2 are nonnegative fictional event counts, so f(x) is their sum. We did not train this extractor on a source dataset; this exercise demonstrates the mechanics of reuse, not measured transfer benefits.

Part	Operation	During this exercise
Input	[count of type 1, count of type 2]	Keep order and units
Extractor	h = max(0, x1 + x2)	Freeze weights and bias
New head	Predict 1 if h >= threshold	Fit the threshold on target training data

The target is a high-activity label. The extractor returns a single number h, not a class label or probability. A new head converts h into the target **prediction**.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Transfer learning trace and record the first step you can explain without help.

Use W2: Fit and compare for **Transfer learning**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Transfer learning explanation against the supplied facts before discussing it.

Use the worked case for **Transfer learning**. Proposed explanation: “Frozen means the whole model cannot learn”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Fit a new head independently

A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

Compare with the **training**-majority baseline, whose tie rule chooses 0.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Trace the reused pipeline

> Trace the reused pipeline. Apply the supplied Transfer learning facts and keep your reasoning.

For **input** [4, 2], compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Fit and compare

> Fit and compare. Apply the supplied Transfer learning facts and keep your reasoning.

For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2.

Explain which data fitted the head.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Diagnose a lost distinction

Diagnose a lost distinction. Apply the supplied Transfer learning facts and keep your reasoning.

For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1.

Does this show that **fine tuning** has been trained and independently validated?

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Fit a new head independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Fit a new head independently. Apply the supplied Transfer learning facts and keep your reasoning.

A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

Compare with the **training**-majority baseline, whose tie rule chooses 0.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Plan credible evidence

> Plan credible evidence. Apply the supplied Transfer learning facts and keep your reasoning.

A report omits the source checkpoint and pre**process**ing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set.

Name the missing records and propose a fair comparison with a model trained from scratch.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Transfer learning to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **transfer learning evidence** sheet to Image-pattern Network Investigation. Show how you can trace a reused **feature extractor** and a newly fitted **output head**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Transfer learning

> Transfer a fixed feature extractor while identifying what is actually refitted. Explain the labelled step before checking the changed condition.

Transfer a fixed **feature extractor** while identifying what is actually refitted.

Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1. Class 1 when h >=threshold; candidates 3 or5.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/28

Transfer learning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Extract features

Training h values are 1, 4, 7.



2. Fit new head

Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5.



3. Check held-out input

Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

CHANGE ONE THING

Query changes to [2, 1].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Transfer a fixed **feature extractor** while identifying what is actually refitted.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

1. Check held-out **input**

2. Extract features

3. Fit new head

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check held-out **input** Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head’s threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Query changes to [2, 1]. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Its h becomes 4 and the fitted head predicts class 0.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its h becomes 4 and the fitted head predicts class 0. Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

Repair a missing building block

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

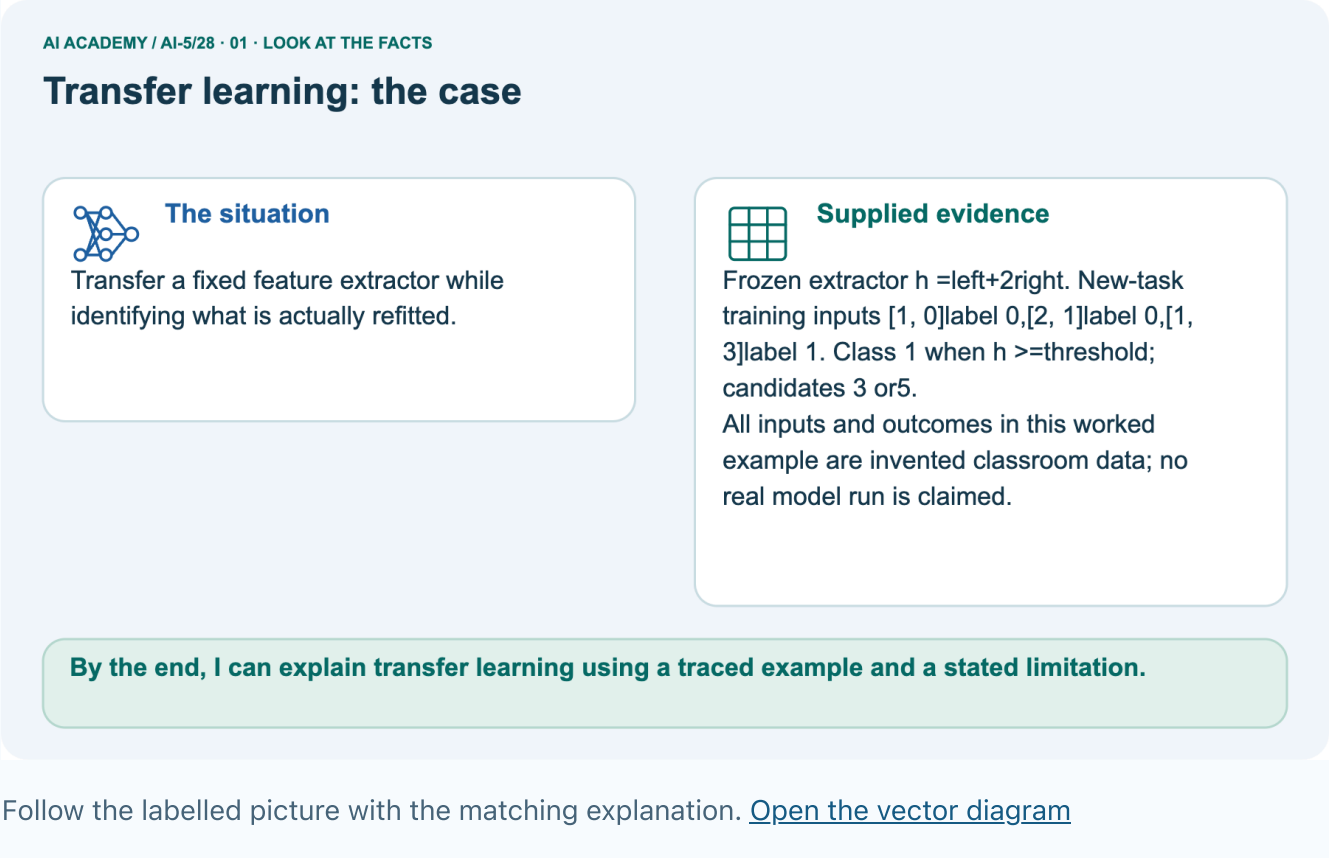
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Transfer learning picture from its labels, then record a relevant limit.

01. Transfer learning: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

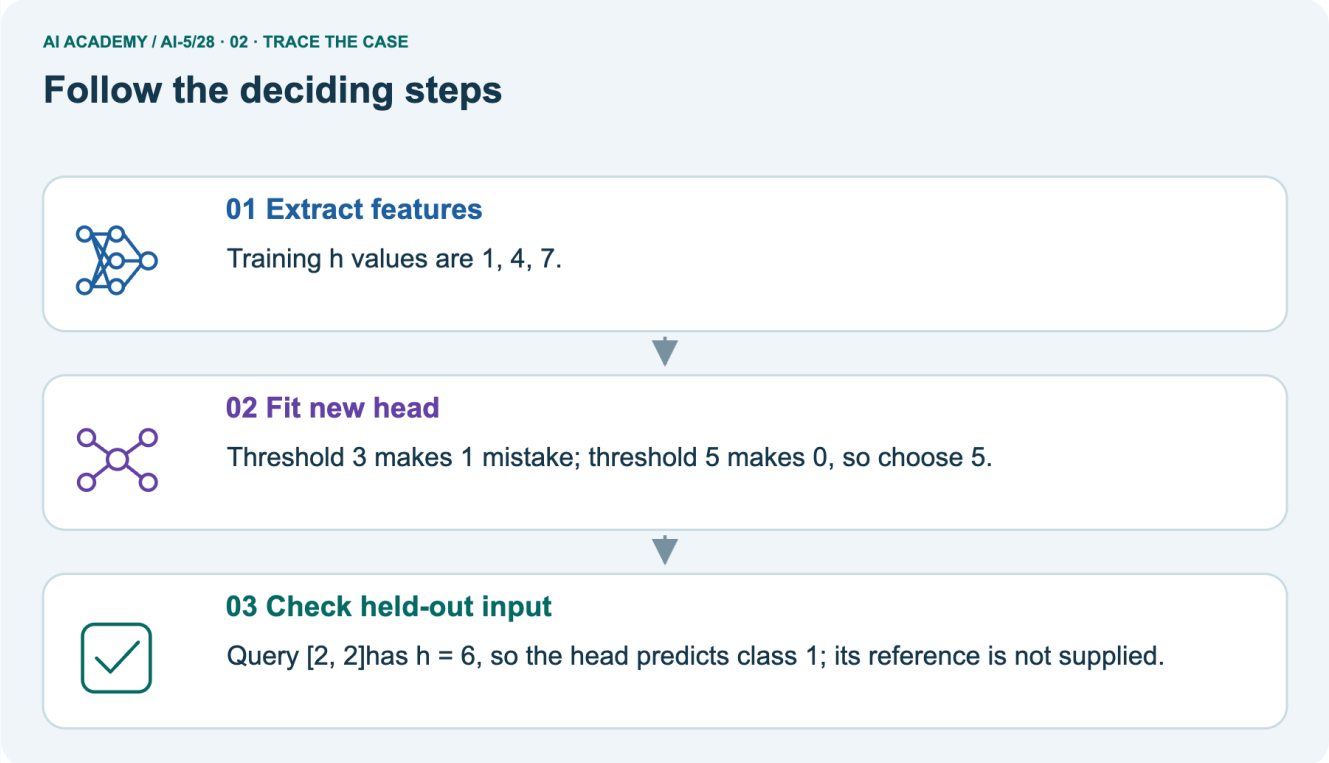
Transfer a fixed **feature extractor** while identifying what is actually refitted. Supplied facts: Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1.

Class 1 when h >=threshold; candidates 3 or5. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Extract features: **Training** h values are 1, 4, 7. → Fit new head: Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5. → Check held-out **input**: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Freeze the fitted head before validation

AI ACADEMY / AI-5/28 · 03 · INSPECT THE LESSON MODEL

Freeze the fitted head before validation

Validation input	Frozen feature	Head at threshold 5
V1 [2, 1], label 0	h = 3	Predict 0, correct
V2 [3, 4], label 1	h = 7	Predict 1, correct

The fitted head is correct on 2 of 2 validation records; the constant -0 baseline is correct on 1 of 2. This tiny constructed comparison is not evidence of general transfer performance.

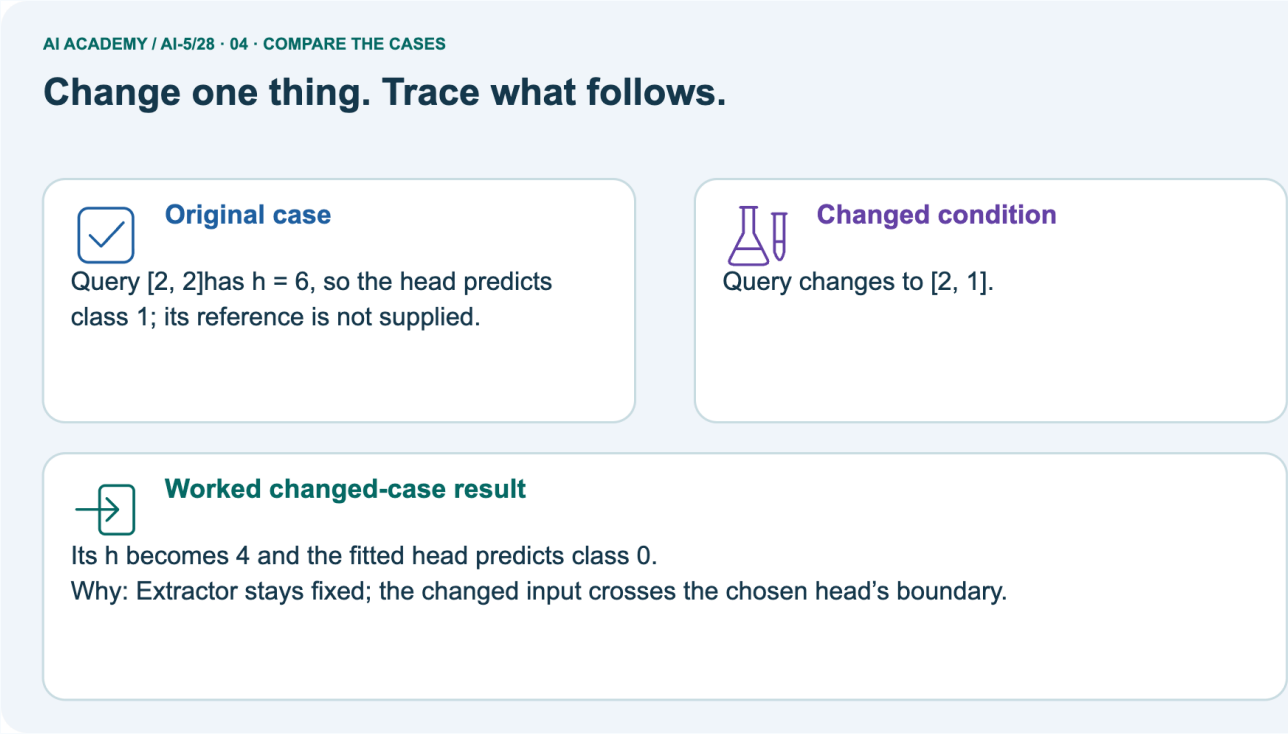
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze the fitted head before validation. V1 [2, 1], label 0 | h = 3 | Predict 0, correct. V2 [3, 4], label 1 | h = 7 | Predict 1, correct

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

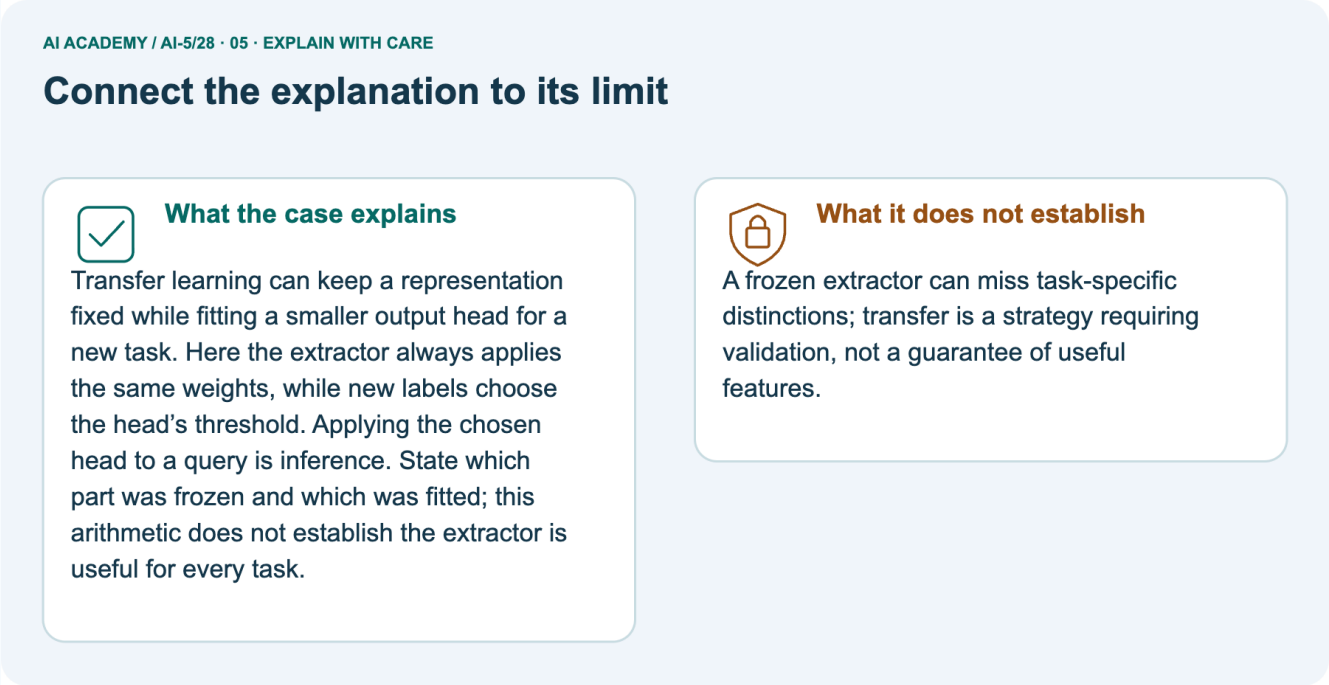
Original case: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied. Changed condition: Query changes to [2, 1]. Expected result: Its h becomes 4 and the fitted head predicts class 0.

Reason: Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

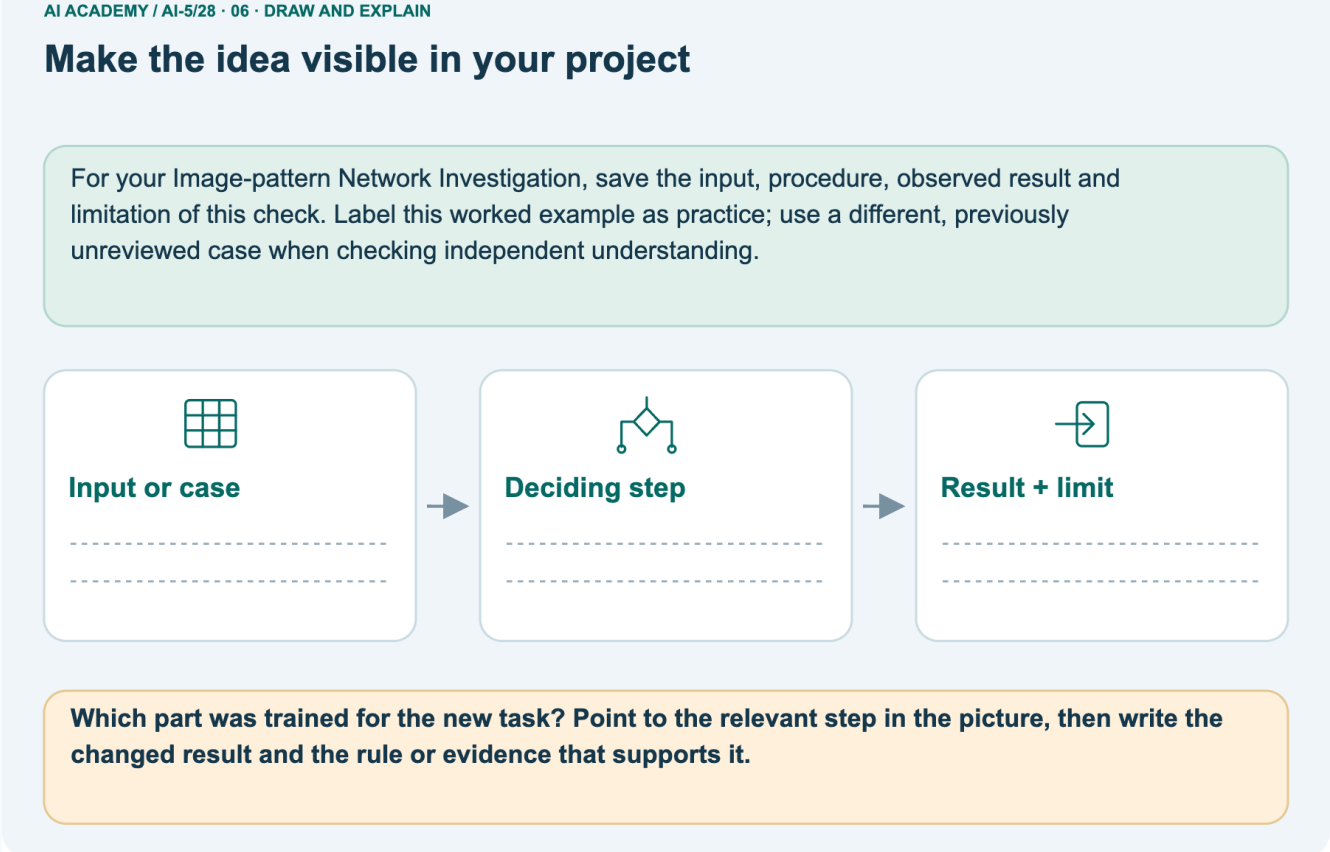
Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Limit: A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring validation, not a guarantee of useful features.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.